EXPERIMENT 1
EXTERNAL FLOW OVER A NACA 0012 AIRFOIL

Computational Fluid Dynamics Laboratory Report

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1. Aim

To study the influence of angle of attack on velocity, pressure, wake and qualitative force trends around a NACA 0012 airfoil at Re = 100. The required angles are −5°, 0°, 5°,10° and 15°.

2. Objectives

- Represent the airfoil on a Cartesian grid using an immersed-geometry mask.
- Use a predictor and pressure-correction procedure to compute the flow field.
- Compare continuity residuals and qualitative lift/drag coefficients.
- Discuss wake and pressure-side/suction-side trends and identify limitations.

3. Numerical method

The supplied notebook creates a Python solver and runs it in a temporary runtime. The airfoil is represented by a smoothed mask χ on a Cartesian mesh. A penalization term suppresses velocity inside the solid region. Convective derivatives use upwind differences; viscous diffusion uses a central-difference Laplacian. A pressure-correction Poisson equation is approximated with Jacobi iterations, after which the velocity is corrected using the pressure-correction gradient. The method is a simplified teaching-level immersed-geometry projection solver.

4. Governing equations

For two-dimensional incompressible flow:

Continuity: $\partial u/\partial x + \partial v/\partial y = 0$

x-momentum: $\partial u/\partial t + u\partial u/\partial x + v\partial u/\partial y = -(1/\rho)\partial p/\partial x + \nu(\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2)$

y-momentum: $\partial v/\partial t + u\partial v/\partial x + v\partial v/\partial y = -(1/\rho)\partial p/\partial y + \nu(\partial^2 v/\partial x^2 + \partial^2 v/\partial y^2)$

Here u and v are velocity components, p is pressure, ρ is density and ν is kinematic viscosity. The supplied implementation advances a velocity predictor with convection, diffusion and mask penalization, then applies a pressure correction to reduce divergence.

5. Geometry, domain and boundary conditions

The NACA 0012 is symmetric, with maximum thickness equal to 12% of chord. The chord is c = 1.0 solver length unit. The domain is Lx = 5.0 and Ly = 2.5, with $x \in [0,5]$ and $y \in [-1.25,1.25]$. The leading-edge reference is x = 1.35; the profile is rotated about the quarter-chord reference.

- Inlet (left): u = 1.0, v = 0.
- Top and bottom: $u = U_\infty$, v = 0.
- Outlet (right): streamwise velocity gradients are approximated as zero; pressure is set to zero.
- Pressure correction: approximate Neumann-style boundary treatment with outlet constraint in the supplied implementation.
- Airfoil: smoothed immersed mask and penalization suppress velocities inside the solid.

These describe the supplied code, not a fully validated production CFD boundary treatment.

6. Parameter set and grid

Parameter	Value
Grid	nx = 350, ny = 350 (custom mode)
Reynolds number	100
Angles	−5°, 0°, 5°, 10°,15°
Free-stream speed	$U_\infty = 1.0$
Chord	c = 1.0
Kinematic viscosity	$\nu = U_\infty c/Re = 0.01$
Domain	5.0 × 2.5
Time step	$\Delta t = 4.201680672269 \times 10^{-3}$
Flow iterations	3000
Poisson/Jacobi iterations per step	60

7. Quantitative comparison

The table reports the final values at iteration 300 from the supplied solver runs using LAB settings. Coefficients are qualitative only.

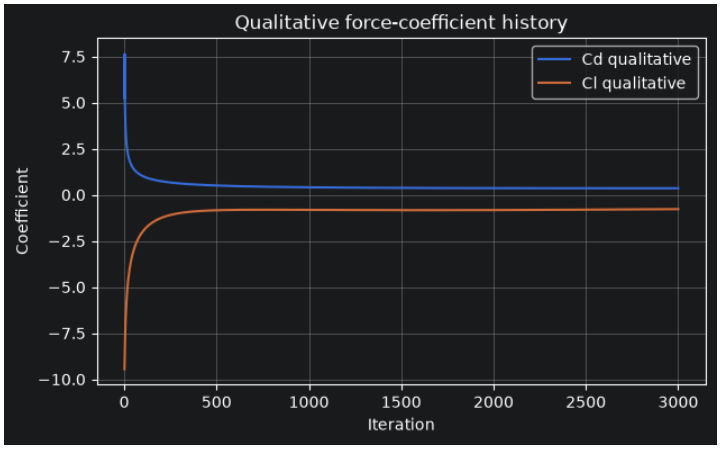
Angle α	Final continuity L2	Maximum speed	Cd qualitative	Cl qualitative
-5°	0.071211	1.196735	0.482040	0.414473
0°	0.070445	1.166777	0.470808	0.000000
5°	0.071211	1.196735	0.482040	-0.414473
10°	0.076469	1.237134	0.448472	-0.745263
15°	0.078932	1.077777	0.525142	-0.776143

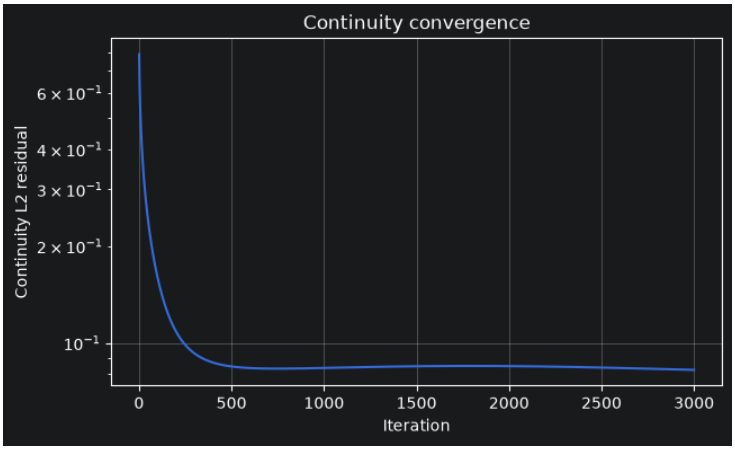
Residual interpretation: the lowest final residual is at 0° (0.070445) and the highest at 15° (0.078932). Residuals drop substantially at first but then plateau. Since they remain non-zero, the cases are not fully converged.

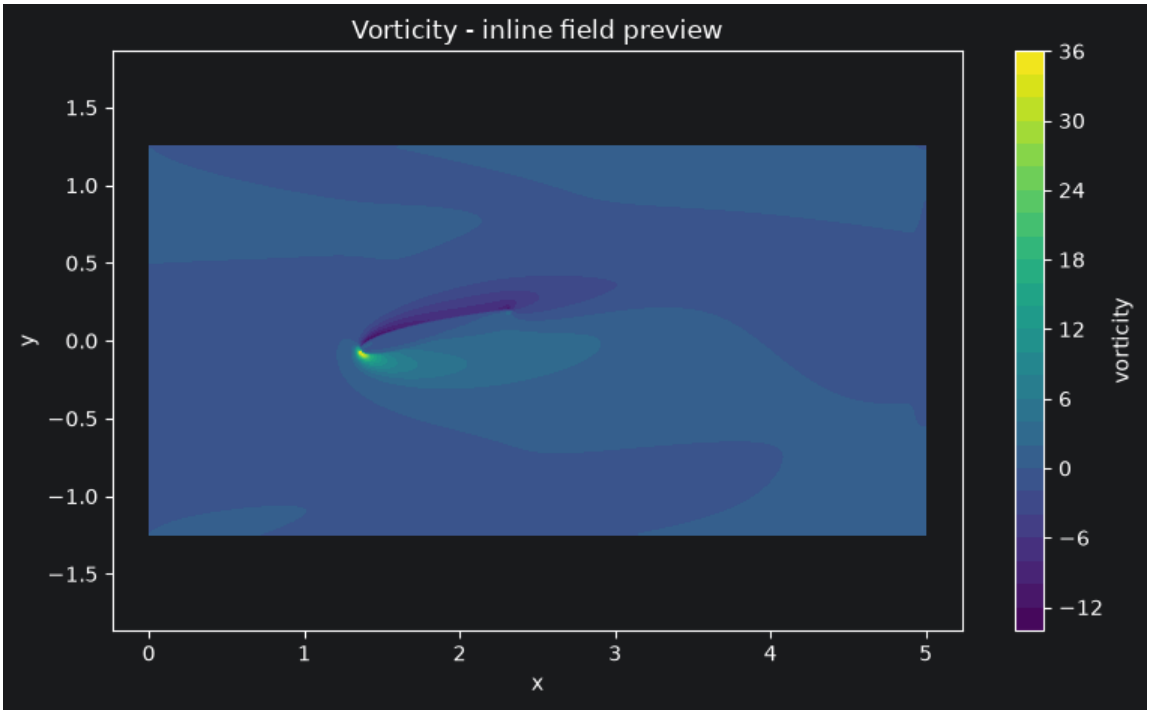
Maximum speed ranges from about 1.1668 to 1.2371. The higher value at 10° indicates stronger local acceleration in this computed field, but it is not a validated physical prediction.

The drag histories tend to decrease toward a plateau. The calculated lift signs for ±5° are opposite under the code's convention to the usual physical expectation. This suggests a sign/convention or implementation issue; do not use these coefficients as validated aerodynamic data.

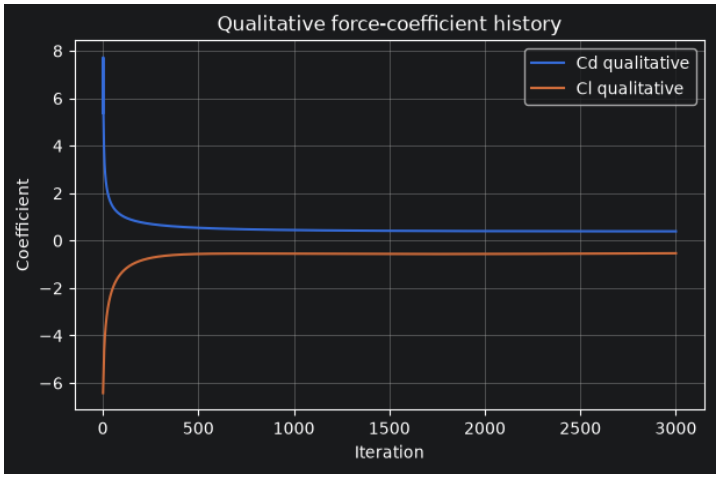
8. Results: AT 15°

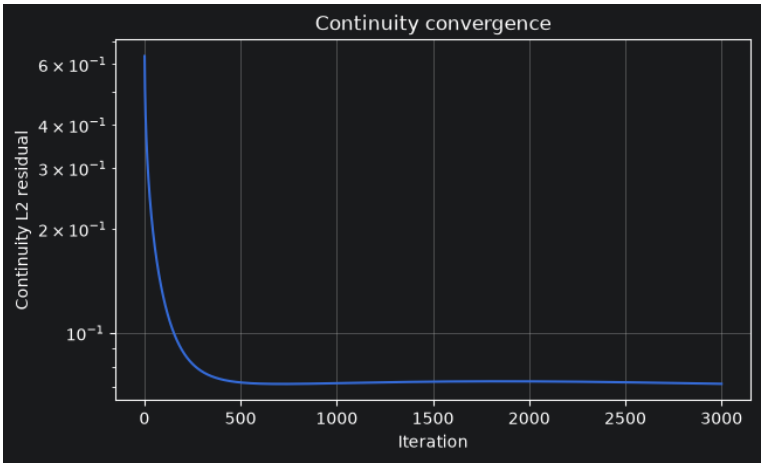


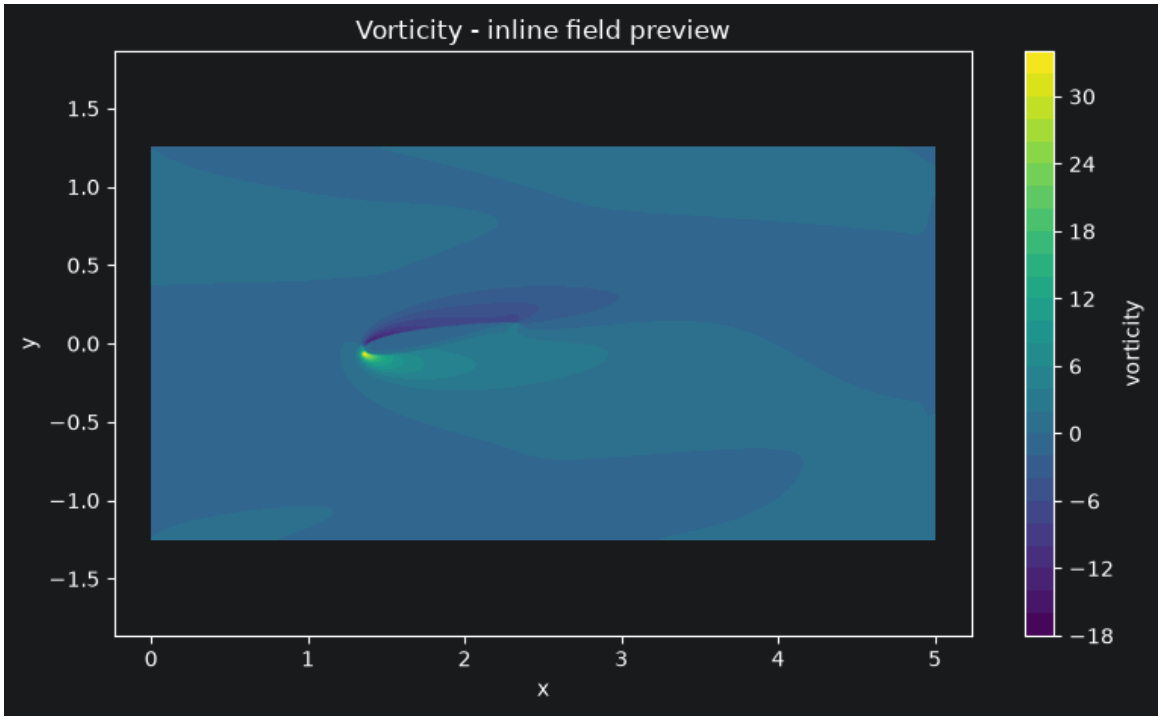



Vorticity - inline field preview
y
x
vorticity
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
1
2
3
4
5
36
30
24
18
12
6
0
−6
−12

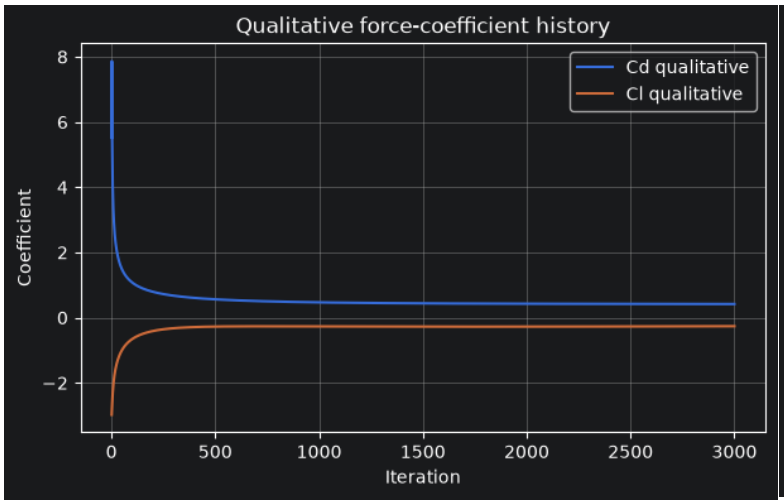
AT 10°

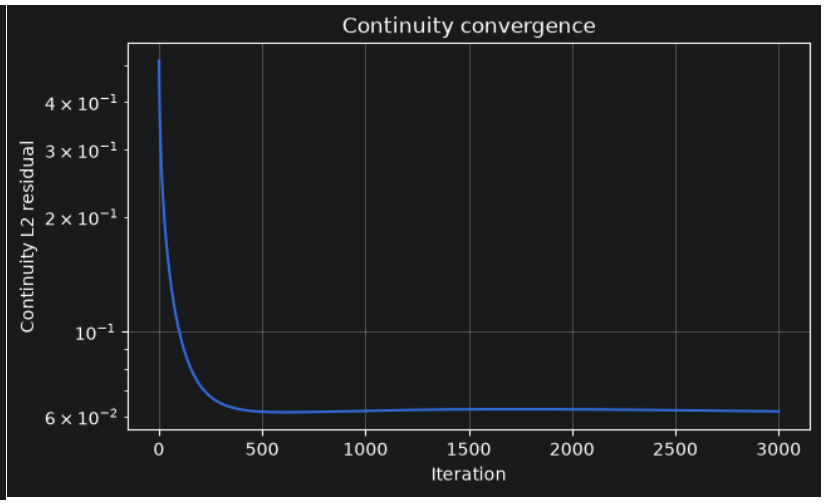

Qualitative force-coefficient history
Cd qualitative
Cl qualitative
Coefficient
Iteration
8
6
4
2
0
−2
−4
−6
0
500
1000
1500
2000
2500
3000

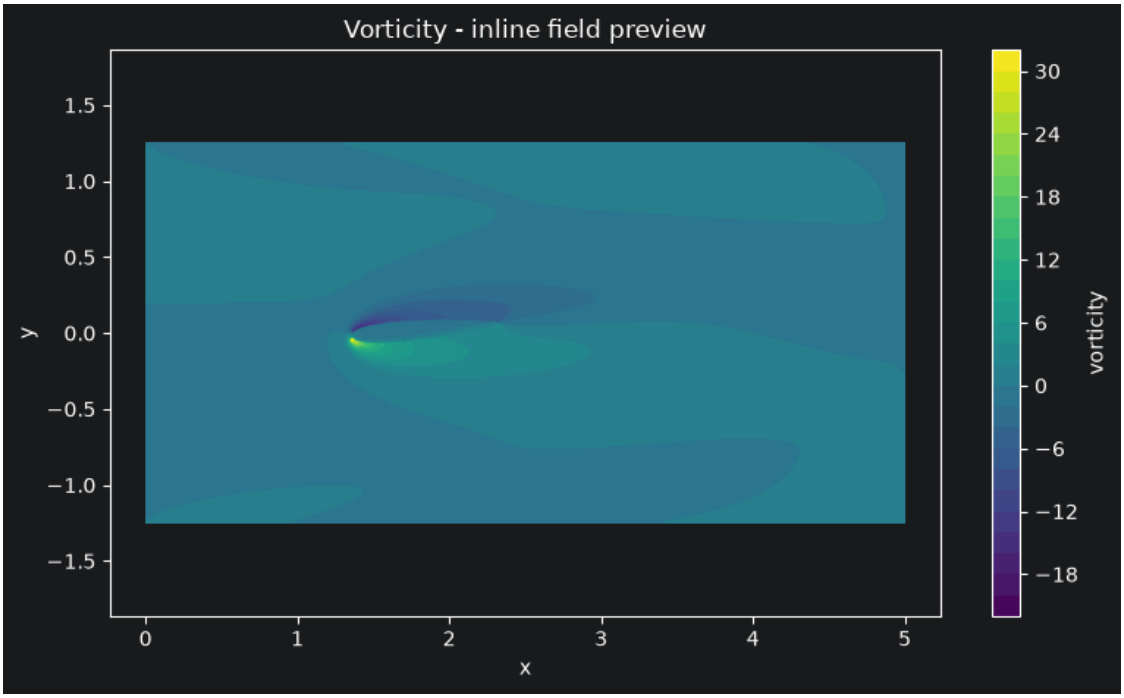

Continuity convergence
Continuity L2 residual
Iteration
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
10^{-1}
0
500
1000
1500
2000
2500
3000


Vorticity - inline field preview
y
x
vorticity
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
1
2
3
4
5
30
24
18
12
6
0
−6
−12
−18

AT 5°


Qualitative force-coefficient history
Cd qualitative
Cl qualitative
Coefficient
Iteration
8
6
4
2
0
−2
0
500
1000
1500
2000
2500
3000


Continuity convergence
Continuity L2 residual
Iteration
4×10^{-1}
3×10^{-1}
2×10^{-1}
10^{-1}
6×10^{-2}
0
500
1000
1500
2000
2500
3000



9. Limitations

- Coarse grid (350 × 350) may blur boundary layers and wake.
- Only 3000 flow iterations are used; continuity residuals remain at a non-zero plateau.
- The pressure Poisson equation uses a fixed 60 Jacobi iterations rather than a strict convergence tolerance.
- The smoothed immersed mask approximates the airfoil boundary.
- No grid-independence study, validated surface Cp, or wall-shear stress is reported.
- Force coefficients are penalization-based and explicitly qualitative.
- Lift-sign behaviour for ±5° appears inconsistent with conventional expectation and should be checked before interpreting forces.

10. Conclusion

The supplied teaching solver was used to compare $\alpha = -5°$, 0°, 5°,10° and 15° at Re = 100. With a 350× 350 grid, 3000 flow iterations and 60 pressure-correction iterations per step, the final continuity residuals ranged from 0.070445 to 0.0789321. The maximum computed speed reached about 1.237 at 10°. The solver produces pressure, speed, vorticity and velocity outputs for visualization, but the residual plateau, coarse mesh, simplified immersed mask and qualitative force estimates limit the conclusions. The lift-sign inconsistency should be noted. The experiment is most useful for studying discretization, pressure correction and convergence rather than reporting high-fidelity aerodynamic coefficients.